Install ESMfold

需要先装 cuda 11.3

```
wget
https://developer.download.nvidia.com/compute/cuda/11.3.0/local_installers/cuda_11
.3.0_465.19.01_linux.run
sudo sh cuda_11.3.0_465.19.01_linux.run
```

等一会会出现一个窗口

输入accept，然后把driver的勾选去掉，"update to this installation" 选 No

最后使用

```
source switch-cuda.sh 11.3
```

切换至cuda 11.3（运行完别忘记切换回11.8）

打开终端，输入

```
conda create -n esm python==3.9 -y
conda activate esm
pip install "fair-esm[esmfold]"
pip install torch==1.12.1+cu113 torchvision==0.13.1+cu113 torchaudio==0.12.1 --
extra-index-url https://download.pytorch.org/whl/cu113
pip install 'dllogger @ git+https://github.com/NVIDIA/dllogger.git'
pip install 'openfold @
git+https://github.com/aqlaboratory/openfold.git@4b41059694619831a7db195b7e0988fc4
ff3a307'
pip install --upgrade numpy==1.26.4
pip install --upgrade pytorch-lightning==1.5
pip install biotite
```

安装完成后，可以使用下面代码验证是否安装成功：

```
import torch
import esm

model = esm.pretrained.esmfold_v1()
model = model.eval().cuda()

# Optionally, uncomment to set a chunk size for axial attention. This can help
reduce memory.
```

```
# Lower sizes will have lower memory requirements at the cost of increased speed.
# model.set_chunk_size(128)

sequence = "MKTVRQERLKSIVRILERSKEPVSGAQLAEELSVSRQVIVQDIAYLRSLGYNIVATPRGYVLAGG"
# Multimer prediction can be done with chains separated by ':'

with torch.no_grad():
    output = model.infer_pdb(sequence)

with open("result.pdb", "w") as f:
    f.write(output)

import biotite.structure.io as bsio
struct = bsio.load_structure("result.pdb", extra_fields=["b_factor"])
print(struct.b_factor.mean())  # this will be the pLDDT
# 88.3
```

可能会OOM (如果出现kill)，需要调大swap：

```
sudo swapoff -a
sudo fallocate -l 16G ./swapfile
sudo chmod 600 ./swapfile
sudo mkswap ./swapfile
sudo swapon ./swapfile
# sudo gedit /etc/fstab
```